# STAT_325_Final

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```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.1     v stringr   1.5.1
## v lubridate 1.9.4     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3

## Loading required package: carData

## Warning: package 'carData' was built under R version 4.4.3

##
## Attaching package: 'car'
##
## The following object is masked from 'package:purrr':
##
##     some
##
## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 4.4.3

## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(MASS)
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
data <- read.csv("C:\\Users\\dilan\\Downloads\\Life Expectancy Data.csv")

data <- data %>%
  drop_na(Life.expectancy, Hepatitis.B, Polio, Diphtheria, GDP, Status, percentage.expenditure)

developing <- data %>%
  filter(Status == "Developing")

developed <- data %>%
  filter(Status == "Developed")
```

```r
summary_stats <- data.frame(
  Variable = c("Life.expectancy", "Hepatitis.B", "Polio", "Diphtheria", "percentage.expenditure"),
  Mean = c(
    mean(developing$Life.expectancy, na.rm = TRUE),
    mean(developing$Hepatitis.B, na.rm = TRUE),
    mean(developing$Polio, na.rm = TRUE),
    mean(developing$Diphtheria, na.rm = TRUE),
    mean(developing$percentage.expenditure, na.rm = TRUE)
  ),
  SD = c(
    sd(developing$Life.expectancy, na.rm = TRUE),
    sd(developing$Hepatitis.B, na.rm = TRUE),
    sd(developing$Polio, na.rm = TRUE),
    sd(developing$Diphtheria, na.rm = TRUE),
    sd(developing$percentage.expenditure, na.rm = TRUE)
  )
)

summary_stats
```

```
##                 Variable      Mean         SD
## 1        Life.expectancy  68.51713   8.185057
## 2            Hepatitis.B  79.64977  25.516433
## 3                  Polio  82.94114  23.161533
## 4             Diphtheria  83.56352  22.185148
## 5 percentage.expenditure 403.11849 910.692819
```

```r
summary_stats2 <- data.frame(
  Variable = c("Life.expectancy", "Hepatitis.B", "Polio", "Diphtheria", "percentage.expenditure"),
  Mean = c(
    mean(developed$Life.expectancy, na.rm = TRUE),
    mean(developed$Hepatitis.B, na.rm = TRUE),
    mean(developed$Polio, na.rm = TRUE),
    mean(developed$Diphtheria, na.rm = TRUE),
    mean(developed$percentage.expenditure, na.rm = TRUE)
  ),
  SD = c(
    sd(developed$Life.expectancy, na.rm = TRUE),
    sd(developed$Hepatitis.B, na.rm = TRUE),
    sd(developed$Polio, na.rm = TRUE),
    sd(developed$Diphtheria, na.rm = TRUE),
    sd(developed$percentage.expenditure, na.rm = TRUE)
  )
)

summary_stats2
```

```
##                 Variable       Mean          SD
## 1        Life.expectancy   79.07509    4.153324
## 2            Hepatitis.B   87.38908   20.763809
## 3                  Polio   94.31058    6.775949
## 4             Diphtheria   93.45392   11.942589
## 5 percentage.expenditure 2481.31858 3370.693067
```

```r
#Developing Model
model1 <- lm(
  Life.expectancy ~ Hepatitis.B + Polio + Diphtheria + percentage.expenditure,
  data = developing
)

summary(model1)
```

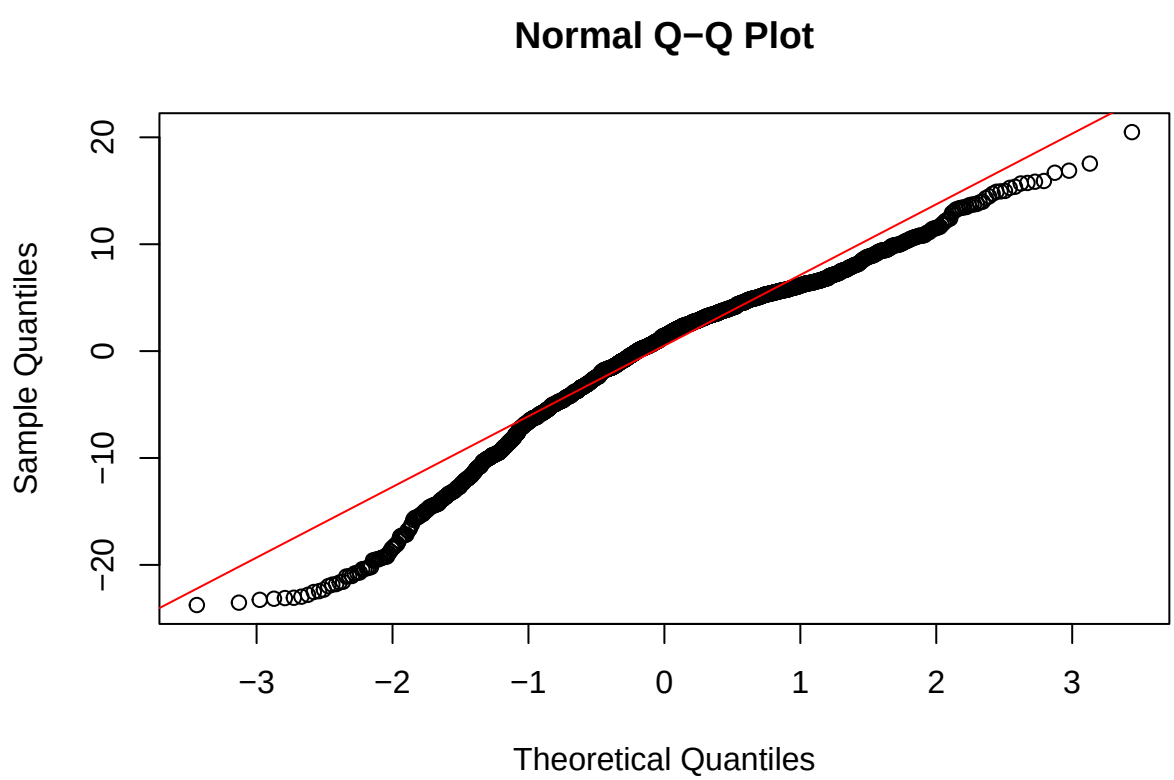
```
##
## Call:
## lm(formula = Life.expectancy ~ Hepatitis.B + Polio + Diphtheria +
##     percentage.expenditure, data = developing)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -23.755  -3.944   1.431   4.973  20.487
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)            5.609e+01  7.367e-01  76.142  < 2e-16 ***
## Hepatitis.B            2.031e-02  9.044e-03   2.246   0.0248 *
## Polio                  5.640e-02  9.529e-03   5.918 3.92e-09 ***
## Diphtheria             5.909e-02  1.138e-02   5.195 2.30e-07 ***
## percentage.expenditure 2.949e-03  1.932e-04  15.261  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.139 on 1711 degrees of freedom
## Multiple R-squared:  0.241,  Adjusted R-squared:  0.2392
## F-statistic: 135.8 on 4 and 1711 DF,  p-value: < 2.2e-16
```

```r
#Check for multicollinearity
vif(model1)
```

```
##            Hepatitis.B                  Polio             Diphtheria
##               1.791889               1.639101               2.143161
## percentage.expenditure
##               1.042195
```

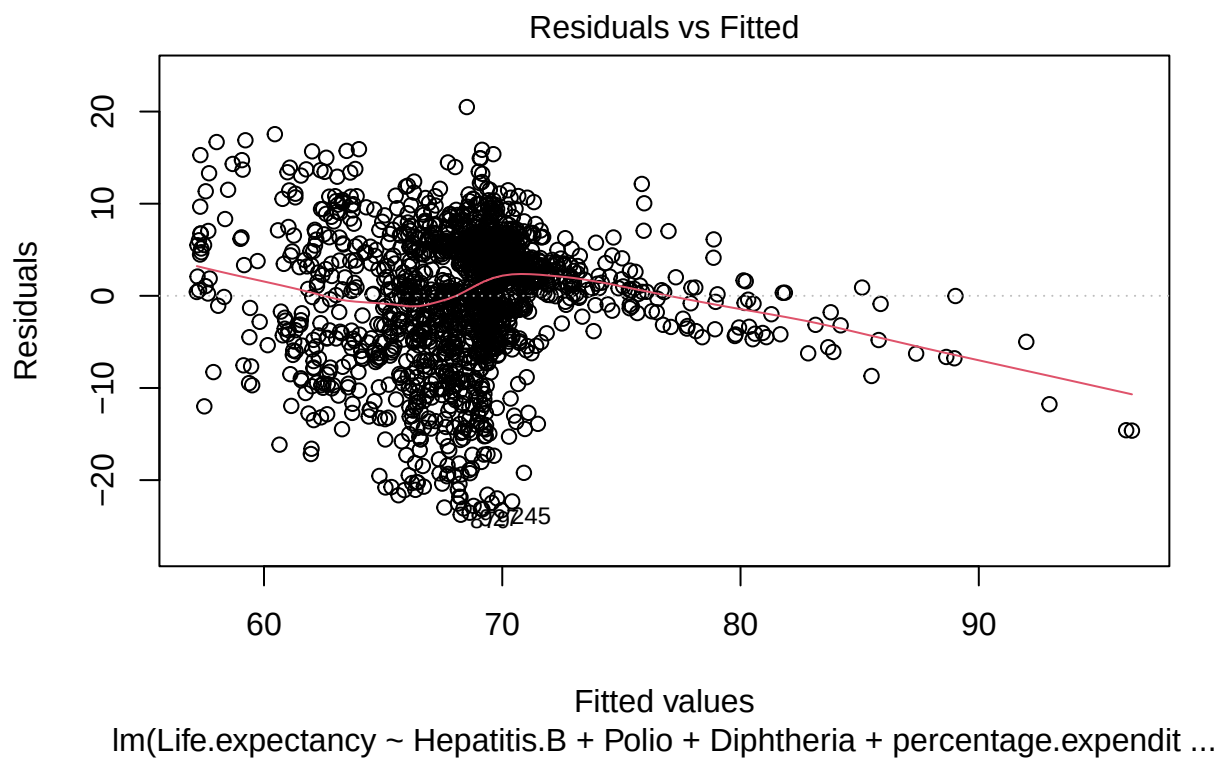
```r
#Normality Assumption
qqnorm(residuals(model1))
qqline(residuals(model1), col = "red")
```



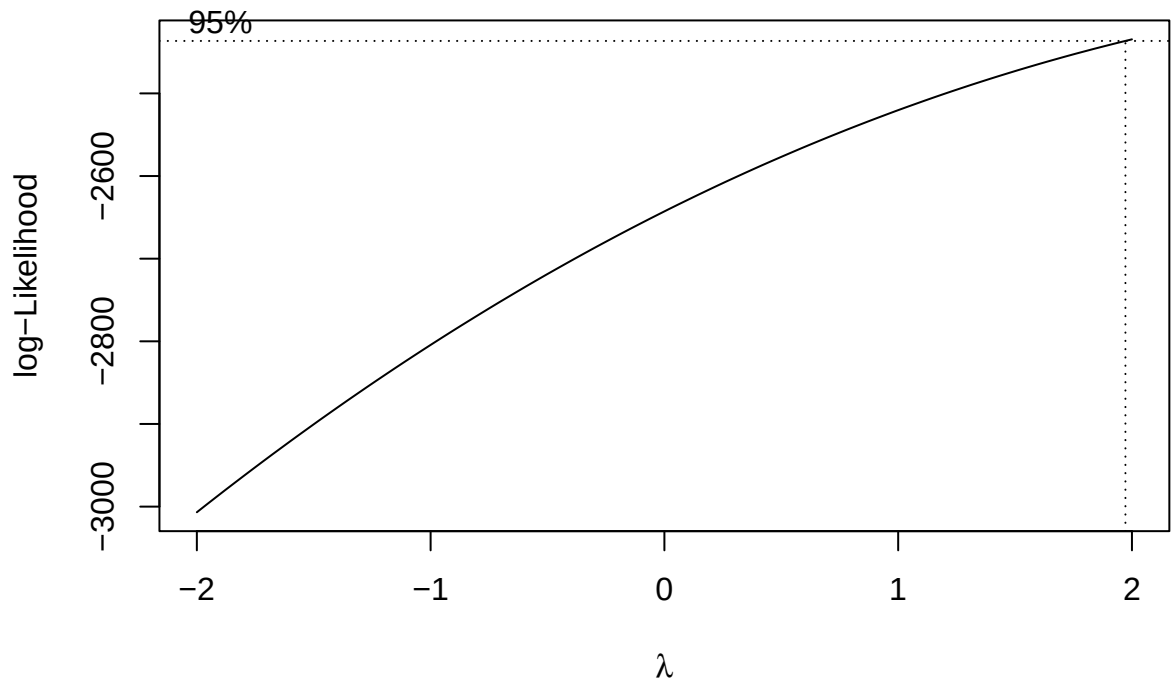
```r
#Constant Varince
bptest(model1)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model1
## BP = 53.053, df = 4, p-value = 8.307e-11
```

```r
plot(model1, which = 1)
```




```r
#BoxCox transformations
boxcox(model1)
```

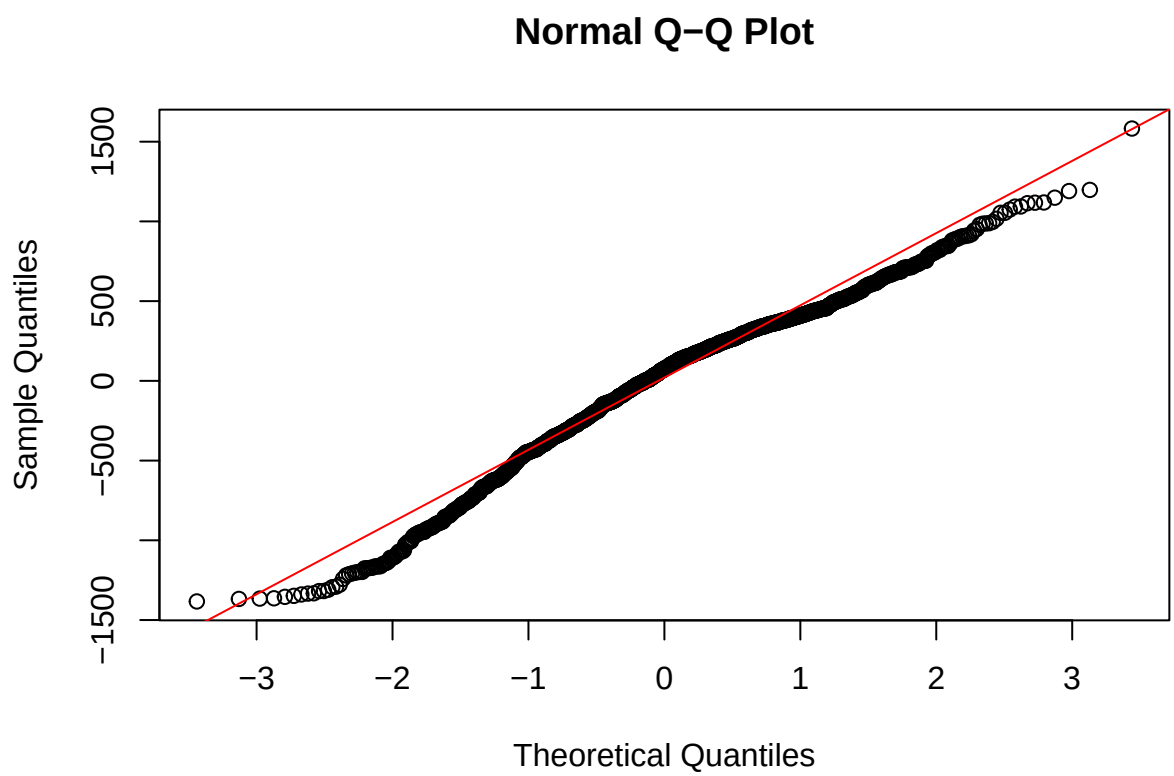



```r
developing$Life.exp.2 <- (developing$Life.expectancy^2 - 1) / 2

model1_lambda2 <- lm(
  Life.exp.2 ~ Hepatitis.B + Polio + Diphtheria + percentage.expenditure,
  data = developing
)

summary(model1_lambda2)
```

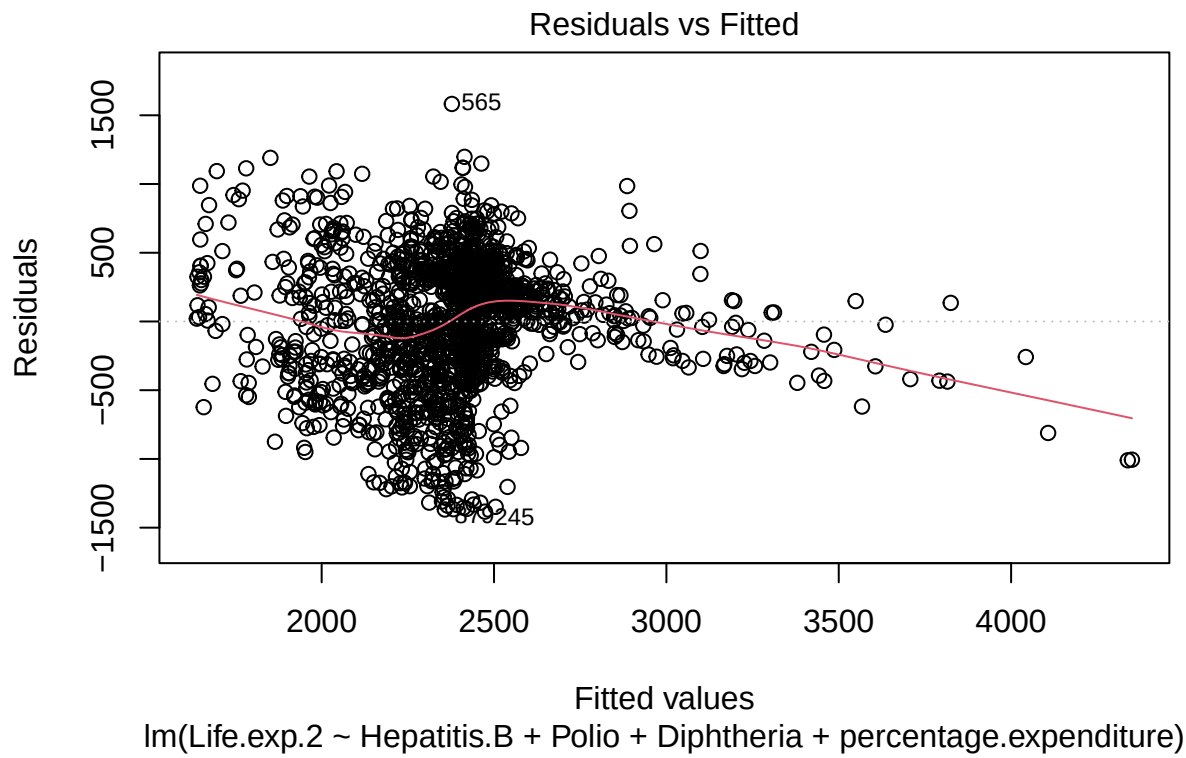
```
##
## Call:
## lm(formula = Life.exp.2 ~ Hepatitis.B + Polio + Diphtheria +
##     percentage.expenditure, data = developing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1383.34  -284.73    75.59   326.30  1582.33
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1567.4448    47.6468  32.897  < 2e-16 ***
## Hepatitis.B               1.2540     0.5849   2.144   0.0322 *
## Polio                     3.6948     0.6163   5.995 2.47e-09 ***
## Diphtheria                3.8601     0.7357   5.247 1.74e-07 ***
## percentage.expenditure    0.2082     0.0125  16.659  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 461.7 on 1711 degrees of freedom
## Multiple R-squared:  0.2588, Adjusted R-squared:  0.257
## F-statistic: 149.3 on 4 and 1711 DF,  p-value: < 2.2e-16
```

```r
#Normality Assumption
# QQ plot
qqnorm(residuals(model1_lambda2))
qqline(residuals(model1_lambda2), col = "red")
```




```r
#Constant Variance
plot(model1_lambda2, which = 1)
```




```r
#Multicollieanrity
vif(model1_lambda2)
```

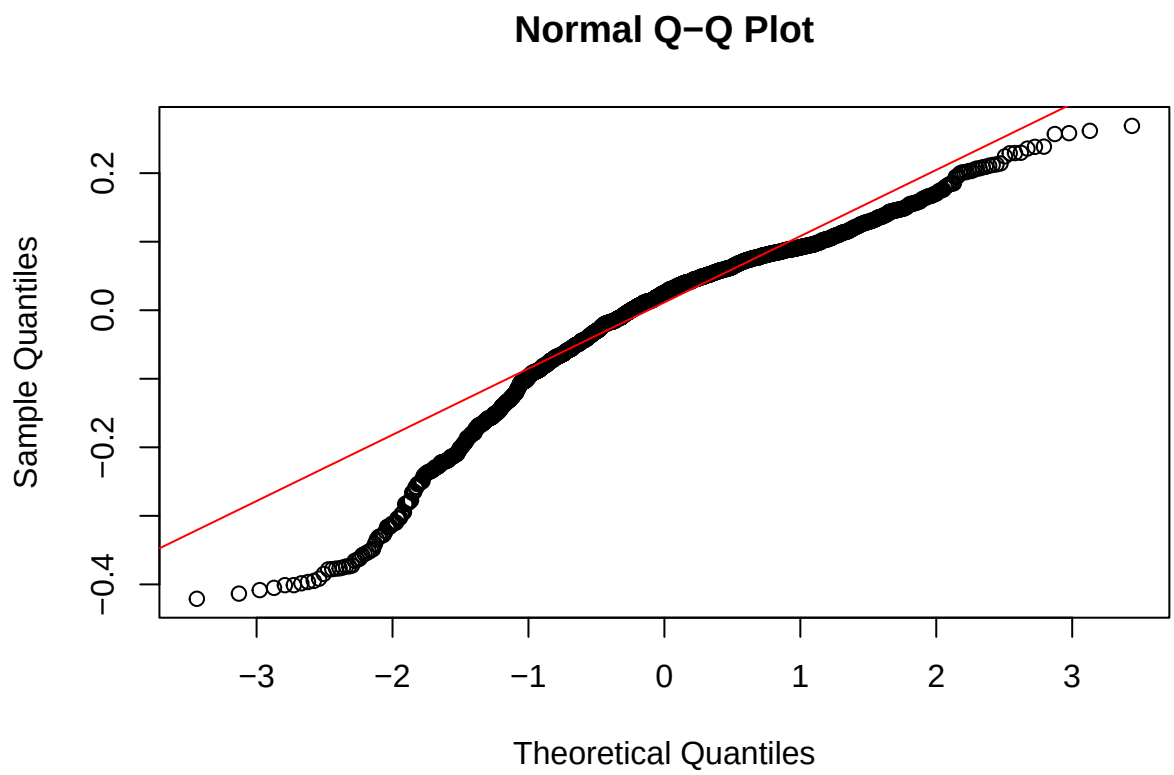
```
##            Hepatitis.B                  Polio             Diphtheria
##               1.791889               1.639101               2.143161
## percentage.expenditure
##               1.042195
```

```r
#Log Transformation
developing$Life.exp.log <- log(developing$Life.expectancy)

model_log <- lm(
  Life.exp.log ~ Hepatitis.B + Polio + Diphtheria + percentage.expenditure,
  data = developing
)

summary(model_log)
```

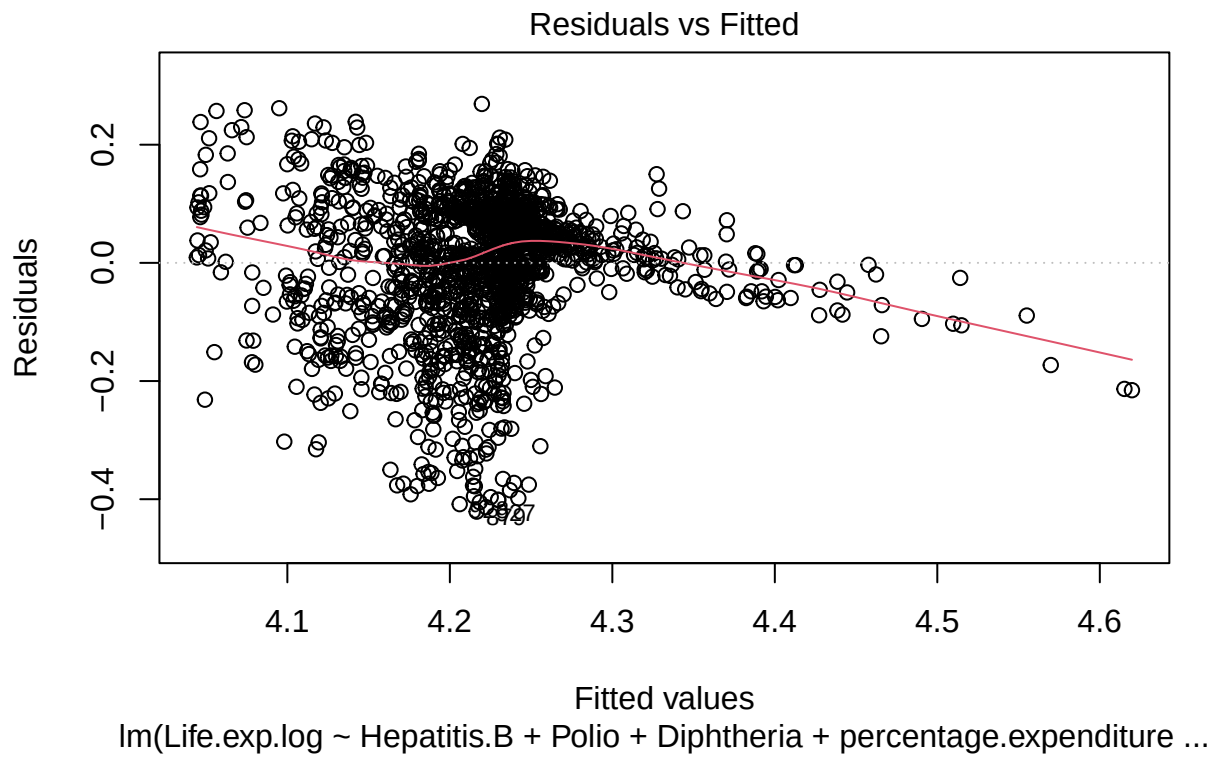
```
##
## Call:
## lm(formula = Life.exp.log ~ Hepatitis.B + Polio + Diphtheria +
##     percentage.expenditure, data = developing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.42101 -0.05371  0.02535  0.07662  0.26894
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.027e+00  1.164e-02 346.055  < 2e-16 ***
## Hepatitis.B            3.307e-04  1.429e-04   2.315   0.0207 *
## Polio                  8.671e-04  1.505e-04   5.760 9.95e-09 ***
## Diphtheria             9.169e-04  1.797e-04   5.102 3.73e-07 ***
## percentage.expenditure 4.232e-05  3.053e-06  13.862  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1128 on 1711 degrees of freedom
## Multiple R-squared:  0.2217, Adjusted R-squared:  0.2199
## F-statistic: 121.9 on 4 and 1711 DF,  p-value: < 2.2e-16
```

```r
#Diagnostics
qqnorm(residuals(model_log)); qqline(residuals(model_log), col="red")
```




```r
plot(model_log, which = 1)
```




```r
vif(model_log)
```

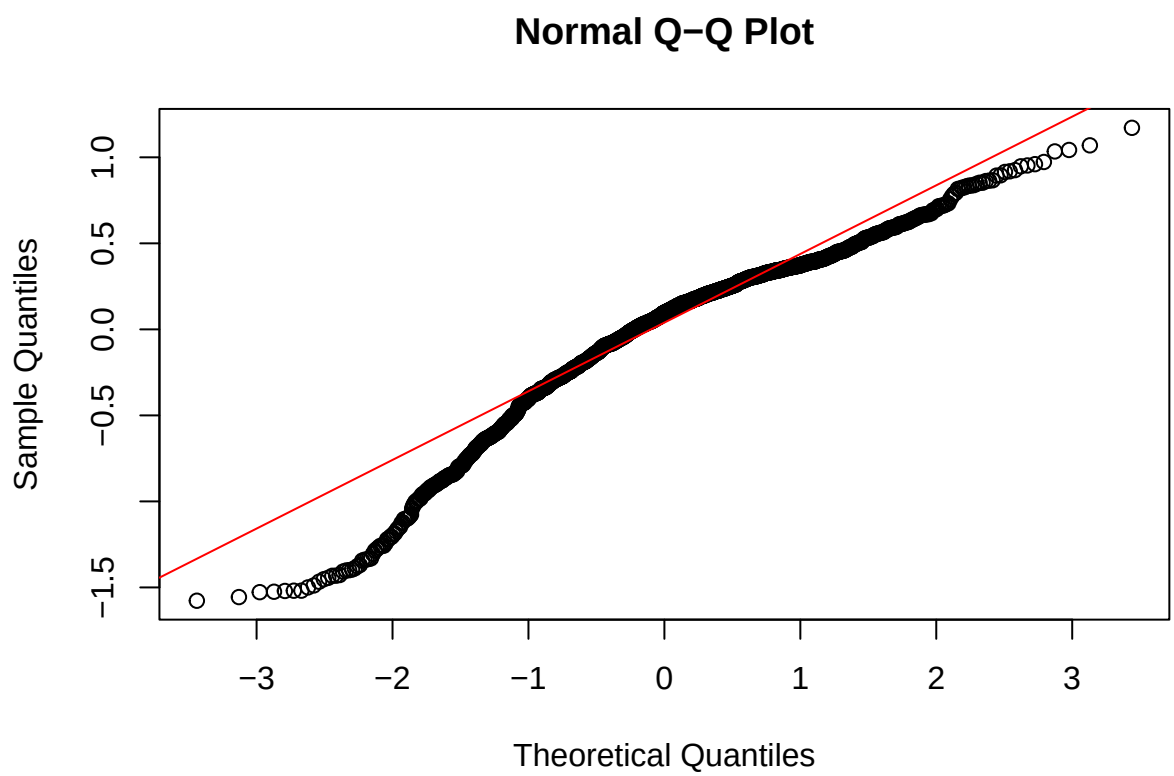
```
##            Hepatitis.B                  Polio             Diphtheria
##               1.791889               1.639101               2.143161
## percentage.expenditure
##               1.042195
```

```r
#Square Transformation
developing$Life.exp.sqrt <- sqrt(developing$Life.expectancy)

model_sqrt <- lm(
  Life.exp.sqrt ~ Hepatitis.B + Polio + Diphtheria + percentage.expenditure,
  data = developing
)

summary(model_sqrt)
```

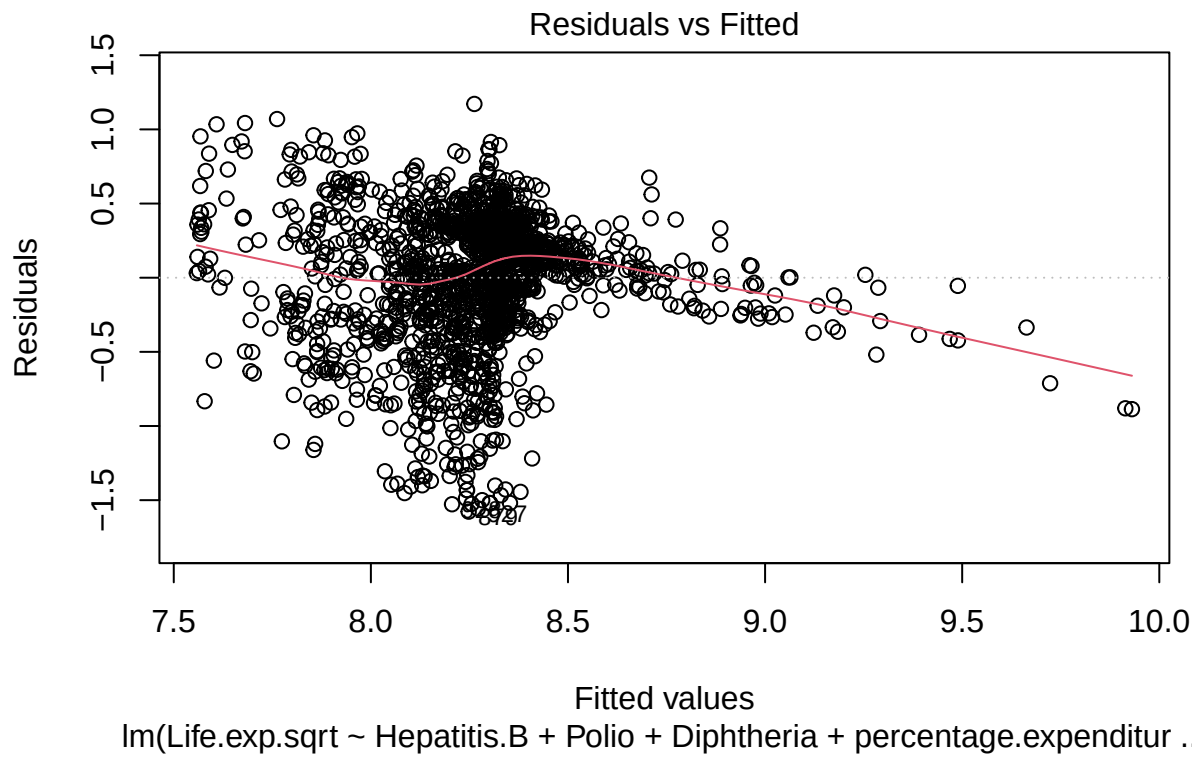
```
##
## Call:
## lm(formula = Life.exp.sqrt ~ Hepatitis.B + Polio + Diphtheria +
##     percentage.expenditure, data = developing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.57726 -0.22989  0.09592  0.30845  1.17146
```

```
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             7.491e+00  4.617e-02 162.246  < 2e-16 ***
## Hepatitis.B             1.295e-03  5.668e-04   2.284   0.0225 *
## Polio                   3.493e-03  5.972e-04   5.850 5.89e-09 ***
## Diphtheria              3.674e-03  7.129e-04   5.154 2.85e-07 ***
## percentage.expenditure 1.763e-04  1.211e-05  14.560  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4474 on 1711 degrees of freedom
## Multiple R-squared:  0.2315, Adjusted R-squared:  0.2297
## F-statistic: 128.9 on 4 and 1711 DF,  p-value: < 2.2e-16
```

```r
#Diagnostics
qqnorm(residuals(model_sqrt)); qqline(residuals(model_sqrt), col="red")
```




```r
plot(model_sqrt, which = 1)
```




```r
vif(model_sqrt)
```

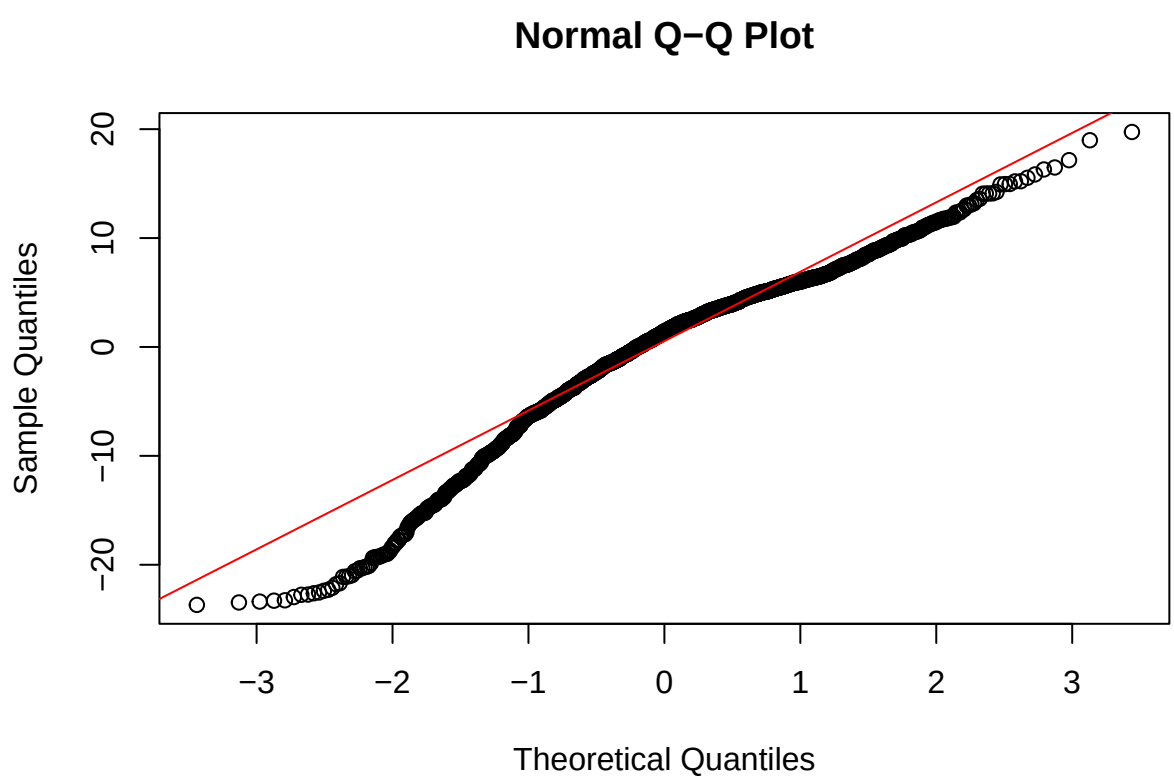
```
##            Hepatitis.B                  Polio             Diphtheria
##               1.791889               1.639101               2.143161
## percentage.expenditure
##               1.042195
```

```r
#Trying Interactions
model_dev_int <- lm(
  Life.expectancy ~
    Hepatitis.B*percentage.expenditure +
    Polio*percentage.expenditure +
    Diphtheria*percentage.expenditure,
  data = developing
)

summary(model_dev_int)
```

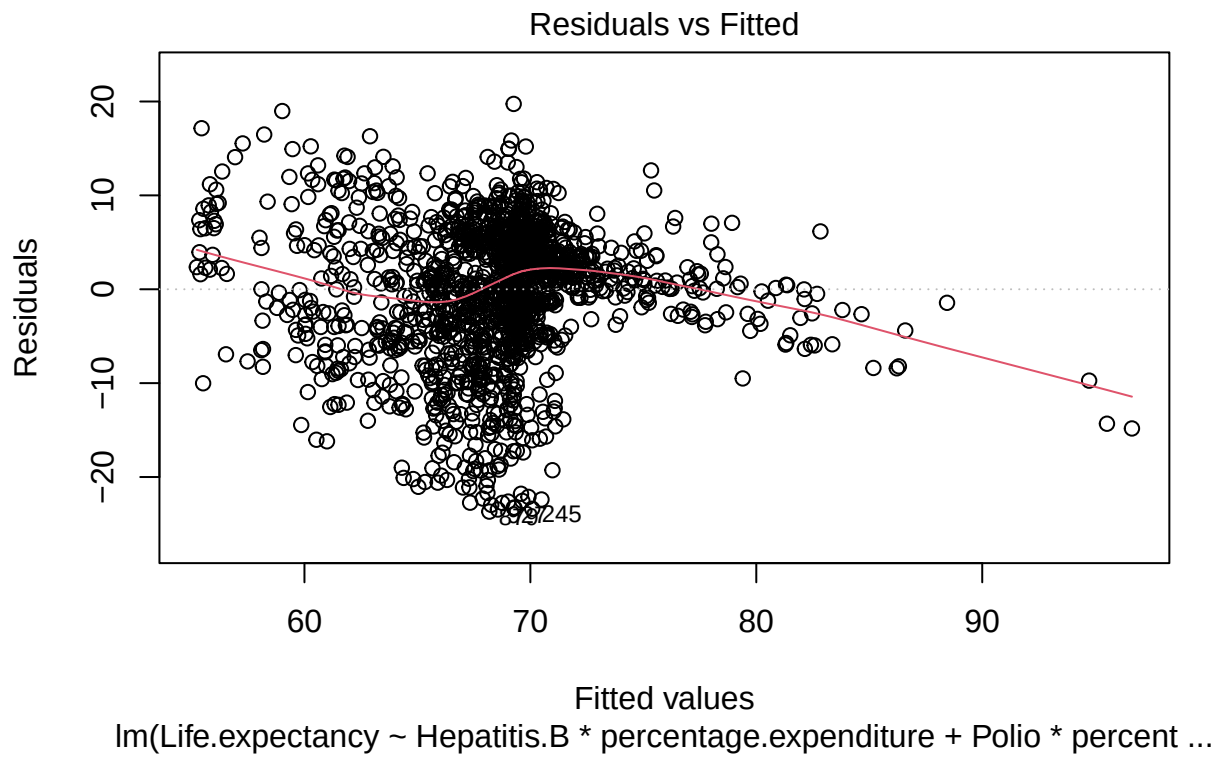
```
##
## Call:
## lm(formula = Life.expectancy ~ Hepatitis.B * percentage.expenditure +
##     Polio * percentage.expenditure + Diphtheria * percentage.expenditure,
##     data = developing)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -23.684  -3.763   1.418   4.834  19.739
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        5.397e+01  8.026e-01  67.250  < 2e-16 ***
## Hepatitis.B                        6.928e-03  9.757e-03   0.710  0.47775
## percentage.expenditure             1.290e-02  1.470e-03   8.779  < 2e-16 ***
## Polio                              6.184e-02  1.003e-02   6.168 8.64e-10 ***
## Diphtheria                         8.935e-02  1.247e-02   7.166 1.15e-12 ***
## Hepatitis.B:percentage.expenditure 1.657e-05  6.026e-06   2.749  0.00604 **
## percentage.expenditure:Polio      -2.469e-05  9.854e-06  -2.505  0.01233 *
## percentage.expenditure:Diphtheria -9.492e-05  1.615e-05  -5.879 4.96e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.034 on 1708 degrees of freedom
## Multiple R-squared:  0.2644, Adjusted R-squared:  0.2614
## F-statistic:  87.7 on 7 and 1708 DF,  p-value: < 2.2e-16
```

```r
#Normality Assumption
# QQ plot
qqnorm(residuals(model_dev_int))
qqline(residuals(model_dev_int), col = "red")
```



```r
#Constant Variance
plot(model_dev_int, which = 1)
```




```r
vif(model_dev_int)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif

##                          Hepatitis.B             percentage.expenditure
##                             2.148147                          62.090524
##                                Polio                         Diphtheria
##                             1.868906                           2.651878
## Hepatitis.B:percentage.expenditure       percentage.expenditure:Polio
##                             5.413318                          24.105960
##  percentage.expenditure:Diphtheria
##                           66.946901
```

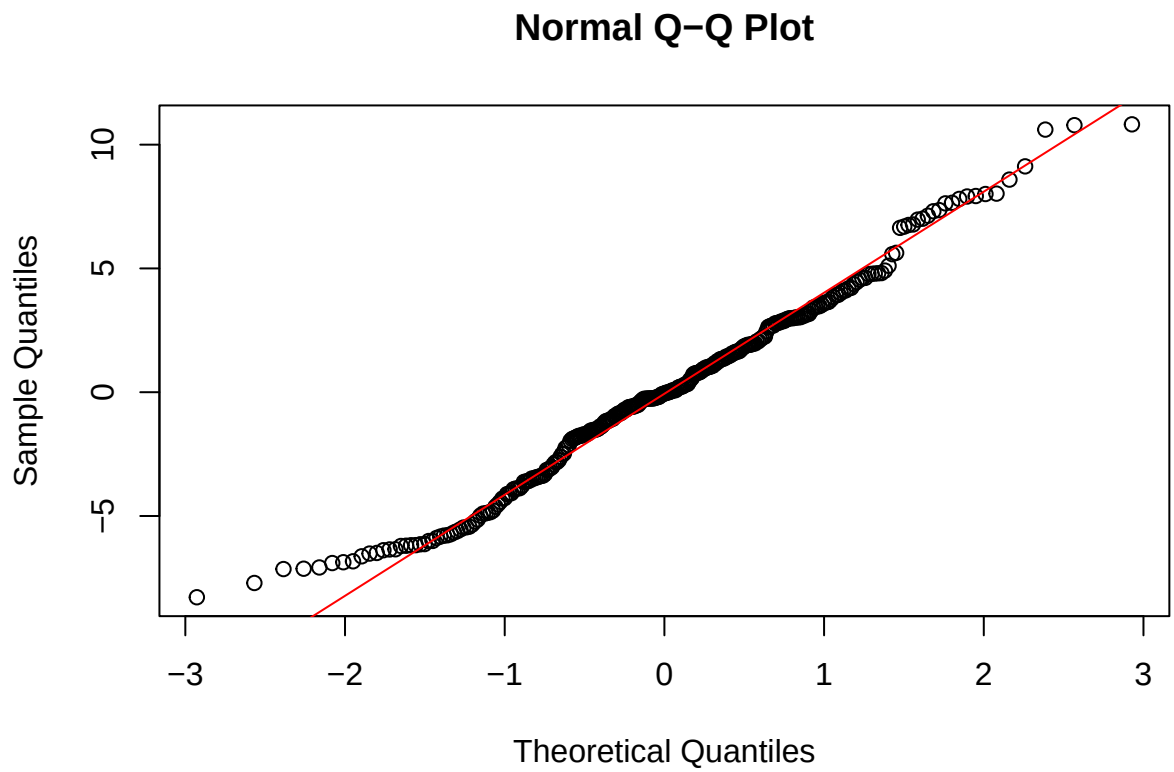
```r
#Developed Model
model2 <- lm(
  Life.expectancy ~ Hepatitis.B + Polio + Diphtheria + percentage.expenditure,
  data = developed
)

summary(model2)
```

```
##
## Call:
## lm(formula = Life.expectancy ~ Hepatitis.B + Polio + Diphtheria +
##     percentage.expenditure, data = developed)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.2817 -2.8148 -0.0527  2.6909 10.8201
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            74.6352043  3.2756940  22.785  < 2e-16 ***
## Hepatitis.B            -0.0105522  0.0113443  -0.930    0.353
## Polio                   0.0453583  0.0364782   1.243    0.215
## Diphtheria              0.0001728  0.0211260   0.008    0.993
## percentage.expenditure  0.0004305  0.0000678   6.349 8.42e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.898 on 288 degrees of freedom
## Multiple R-squared:  0.1314, Adjusted R-squared:  0.1193
## F-statistic: 10.89 on 4 and 288 DF,  p-value: 3.083e-08
```

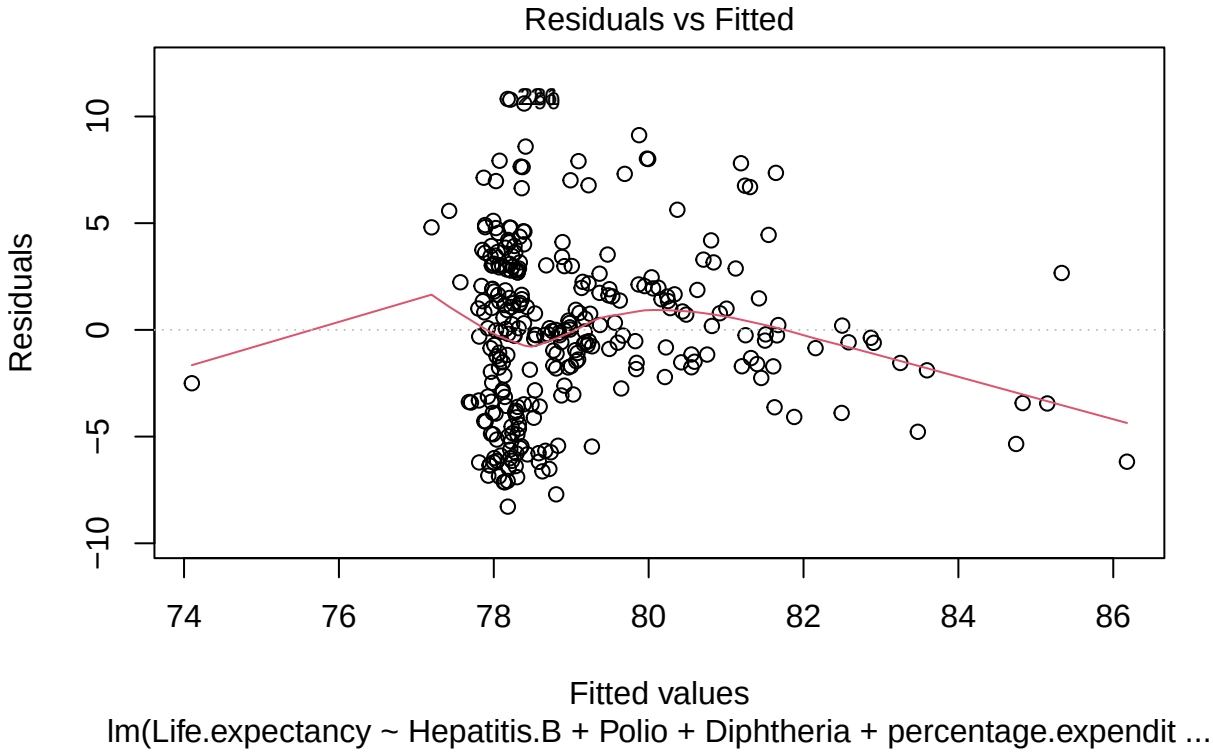
```r
#Multicollinearity
vif(model2)
```

```
##            Hepatitis.B                   Polio              Diphtheria
##               1.066487                1.174330                1.223526
## percentage.expenditure
##               1.003901
```

```r
#Normality
qqnorm(residuals(model2))
qqline(residuals(model2), col = "red")
```




```r
#Constant variance
plot(model2, which = 1)
```


Residuals vs Fitted
Residuals
Fitted values
lm(Life.expectancy ~ Hepatitis.B + Polio + Diphtheria + percentage.expendit ...
10
5
0
−5
−10
74
76
78
80
82
84
86